

Data Cloud Semantic Search Migration

GetCaseSummaryFields + FindSimilarCases

Uphold FinTech — Agentforce POC

This document covers all five deliverables for migrating the Uphold RFI Agent's Apex actions from SOQL/SOSL keyword matching to Data Cloud semantic vector search. Response types (CaseSummaryResponse, SimilarCasesResponse, etc.) are unchanged.

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1. Data Cloud Setup Checklist

1.1 CRM Data Streams

Create three CRM Data Streams to ingest Salesforce objects into Data Cloud. Navigate: **Setup** → **Data Cloud** → **Data Streams** → **New** → **Salesforce CRM**

Object	Stream Name	Category	Key Fields to Include
Case	Case_Home	Other	Id (PK), CaseNumber, Subject, Description, Status, Priority, Origin, CreatedDate, Clo
RFI_Question__c	RFI_Question__c_Home	Other	Id (PK), Name, Question_Text__c, Response_Text__c, Category__c, Status__c, Pric
RFI_Approval__c	RFI_Approval__c_Home	Other	Id (PK), Case__c (FK), Approver__c, Status__c, Comments__c, Approved_Date__c,

1.2 DMO Field Mapping

After each data stream finishes processing, map it to a Data Model Object. Navigate to the stream → **Data Mapping** → **Start** → **Select Objects**.

- **Case:** Auto-maps to standard **ssot__Case__dlm**. Verify 10 fields.
- **RFI_Question__c:** Creates custom DMO. Map 12 fields — **critically include Case__c** (FK to parent Case).
- **RFI_Approval__c:** Creates custom DMO. Map 8 fields — include Case__c (FK).

Primary Keys and Foreign Keys

Each DMO must have Id marked as Primary Key. The Case__c lookup field on RFI_Question__c and RFI_Approval__c must be mapped

1.3 Search Index Creation

Navigate: **Setup** → **Data Cloud** → **Search Indexes** → **New**

Setting	Case Index	RFI Question Index (Optional)
Setup Type	Advanced Setup	Advanced Setup
Search Type	Hybrid Search	Hybrid Search
Source Object	Case (DMO)	RFI_Question__c (custom DMO)
Fields to Chunk	Subject, Description	Question_Text__c, Response_Text__c
Embedding Model	E5 Large V2	E5 Large V2
Chunk Size	Default (8000 tokens)	Default (8000 tokens)
Overlap	Default (512 tokens)	Default (512 tokens)
Result DMO Names	Case_index__dlm Case_chunk__dlm	RFI_Question__c_index__dlm RFI_Question__c_chunk__dlm

The search index is explicitly created — it is NOT auto-generated when you embed fields. Wait until status shows **Ready** before testing queries.

1.4 Test in Query Editor

Navigate: **Data Cloud** → **Query Editor** → **New**. Run this query replacing the search phrase:

```
SELECT v.score__c AS Score, c.ssot__Id__c AS CaseId, c.ssot__Subject__c AS Subject
FROM vector_search(TABLE(Case_index__dml), 'account restriction order', '', 5) v
JOIN Case_chunk__dml cc ON v.RecordId__c = cc.RecordId__c
JOIN ssot__Case__dml c ON cc.SourceRecordId__c = c.ssot__Id__c
ORDER BY Score DESC LIMIT 5
```

1.5 Permission Sets

Permission	Where	Required For
Data Cloud User or Data Cloud Admin	Setup → Permission Sets → Assign to agent user	ConnectApi.CdpQuery access in Apex
Apex class API version ≥ 59.0	Class metadata (.cls-meta.xml)	ConnectApi namespace availability
CRM Connector active	Setup → Data Cloud → Salesforce CRM	Data ingestion from Salesforce objects

2. Updated GetCaseSummaryFields.cls

Changes Summary

What Changed	Before	After
@InvocableMethod	No callout flag	callout=true added
Strategy 3 (keyword search)	SOQL Subject LIKE + multi-keyword AND	ConnectApi.CdpQuery vector_search on Case_index__dlm
Strategy 4 (SOSL)	Primary fallback for keyword search	Demoted to final safety net (fires only if Data Cloud fails)
New method	—	findCaseBySemanticSearch() private helper
SOQL hydration	N/A (direct SOQL)	After vector_search returns Case ID, SOQL fetches full record + relationships

Search Flow

1. **Record ID?** → Direct SOQL (unchanged)
2. **Case Number?** → Direct SOQL (unchanged)
3. **Keywords?** → **Data Cloud vector_search** (returns best-match Case ID) → SOQL hydration
4. **Data Cloud fails/empty?** → SOSL safety net (unchanged)

Key Code: findCaseBySemanticSearch()

```
ConnectApi.CdpQueryInput cdpInput = new ConnectApi.CdpQueryInput();
cdpInput.sql = 'SELECT v.score__c AS Score, c.ssot__Id__c AS CaseId '
+ 'FROM vector_search(TABLE(Case_index__dlm), \''
+ String.escapeSingleQuotes(keywords) + '\', \'\', 1) v '
+ 'JOIN Case_chunk__dlm cc ON v.RecordId__c = cc.RecordId__c '
+ 'JOIN ssot__Case__dlm c ON cc.SourceRecordId__c = c.ssot__Id__c '
+ 'ORDER BY Score DESC LIMIT 1';

ConnectApi.CdpQueryOutput cdpOutput = ConnectApi.CdpQuery.queryANSISql(cdpInput);

// Extract Case ID → hydrate via standard SOQL
Map<String, Object> topRow = (Map<String, Object>) JSON.deserializeUntyped(
    JSON.serialize(cdpOutput.data[0]));
String caseId = String.valueOf(topRow.get('CaseId'));
// → SELECT ... FROM Case WHERE Id = :caseId LIMIT 1
```

■ Why SOQL Hydration?

Data Cloud returns only indexed DMO fields (Subject, Description, Id). The Case summary needs Account.Name, Contact.Name, and all

Unchanged:

- @InvocableMethod label, Request/Response classes, CaseSummaryResponse shape
- Strategy 1 (Record ID) and Strategy 2 (Case Number) — exact lookups, no search needed
- RFI_Question__c child record retrieval (standard SOQL after Case is found)

3. Updated FindSimilarCases.cls

Changes Summary

What Changed	Before	After
@InvocableMethod	No callout flag; generic description	callout=true; description mentions semantic search
Strategy 1 (keyword search)	SOQL Subject LIKE single-pattern match	ConnectApi.CdpQuery vector_search returns ranked list
Strategy 2 (SOSL)	Primary fallback	Safety net only (fires if Data Cloud fails)
New method	—	findCasesBySemanticSearch() returns List<Case>
Result ordering	CreatedDate DESC	Semantic similarity score DESC (preserved through hydration)

Key Difference from GetCaseSummaryFields

FindSimilarCases returns **up to N results** (default 10) ranked by similarity score, whereas GetCaseSummaryFields returns **exactly 1** (the best match). The vector_search LIMIT is set to **limitVal** instead of 1.

Rank-Order Preservation

Data Cloud returns results ordered by score__c DESC. After SOQL hydration (which returns results in arbitrary order), the code **reconstructs the original rank order** by iterating through the saved caseIds list and looking up each in the SOQL Map:

```
List<String> caseIds = new List<String>(); // preserves DC rank order
for (Object rowObj : cdpOutput.data) { ... caseIds.add(caseId); }

Map<Id, Case> caseMap = new Map<Id, Case>([SELECT ... WHERE Id IN :caseIds]);

// Reconstruct in semantic-rank order
List<Case> orderedResults = new List<Case>();
for (String cid : caseIds) {
    if (caseMap.containsKey(cid)) orderedResults.add(caseMap.get(cid));
}
```

Unchanged:

- @InvocableMethod label = 'Find Similar RFI Cases'
- Request/Response inner classes and SimilarCasesResponse shape
- limitVal cap of 10 results
- SimilarCaseItem field mapping (caseNumber, subject, status, etc.)

4. Field Embedding Recommendation

Which fields should feed the embedding input for the Case search index? The answer depends on the compliance RFI use case.

Option	Fields	Pros	Cons	Verdict
A: Subject only	Subject	• Clean, concise signal • Fast indexing • Low noise	• Misses regulatory detail in Description • Many Subjects are generic ("Q4 request")	Too narrow for compliance RFI
B: Subject + Description	Subject, Description	• Captures regulatory context • Description has entity names, citation refs, procedure details • Standard DC Search Index pattern (no custom work)	• Description can be noisy (boilerplate, signatures) • Longer chunks but E5 handles up to 8K tokens well	■ RECOMMENDED for initial rollout
C: Subject + RFI Questions	Subject + all child Question_Text_*	• Most semantically rich • Matches coordinator shorthand to regulator vocabulary • Best theoretical recall	• Requires multi-object embed (calculated field or Apex) • Index rebuild on every child record change • Significantly more complex	Phase 2 upgrade (after B is proven)

■ Recommendation: Start with Option B (Subject + Description)

For a compliance RFI use case, Description contains the regulatory detail that makes semantic search valuable — entity names (IRS, SEC, etc.)

Trade-off Analysis for Compliance RFI

In compliance work, the vocabulary gap is the core problem:

- **Coordinator types:** "freeze customer account"
- **Historical case says:** "account restriction order pursuant to BSA/AML hold"
- **RFI question says:** "Describe the process for implementing a custodial account restriction under 31 CFR 1010.610"

With **Subject + Description**, the embedding model (E5 Large V2) encodes the regulatory vocabulary from the Description field, enabling semantic matching even when zero lexical tokens overlap between the coordinator's query and the historical record.

5. Deployment Notes

Metadata Artifacts to Change

Artifact	Change	Why
GetCaseSummaryFields.cls	Replace class body (full rewrite of findCase())	Semantic search + SOQL hydration + SOSL safety net
GetCaseSummaryFields.cls-meta.xml	Set apiVersion to 62.0	ConnectApi.CdpQuery requires API v59.0+; target is v62.0
FindSimilarCases.cls	Replace class body (new findCasesBySemanticSearch())	Semantic search + rank-order preservation + SOSL safety net
FindSimilarCases.cls-meta.xml	Set apiVersion to 62.0	Same as above
Permission Set: Data_Cloud_Agent_User (new or existing)	Assign Data Cloud User to Agentforce running user	ConnectApi.CdpQuery requires Data Cloud permission

What Does NOT Change

- **CaseSummaryResponse.cls** — unchanged (CLT response type)
- **RfiQuestionResponse.cls** — unchanged (nested CLT type)
- **SimilarCasesResponse.cls** — unchanged
- **SimilarCaseItem.cls** — unchanged
- **Agent Script / Bot metadata** — no re-registration needed (same @InvocableMethod labels)
- **CLT components (LWC)** — unchanged (same response shapes)

What Does NOT Need to Be Created

- **Named Credential** — NOT needed. ConnectApi.CdpQuery is a server-side API, not an HTTP callout. The callout=true flag is required because the ConnectApi internally makes a callout to the Data Cloud service.
- **Connected App** — NOT needed. Data Cloud uses the org's built-in CRM Connector.
- **Custom Metadata** — NOT needed. Search index names are hardcoded in the Apex (Case_index__dlm, Case_chunk__dlm). If you want them configurable, create a Custom Metadata Type later.
- **Remote Site Setting** — NOT needed. ConnectApi handles its own connectivity.

Deployment Order

1. Complete Data Cloud setup (Section 1) — streams, mapping, search index, verify Ready status
2. Assign Data Cloud User permission set to the Agentforce running user
3. Deploy updated .cls and .cls-meta.xml files for both classes
4. Test in Query Editor first, then via Agentforce agent
5. Verify SOSL fallback works by temporarily disabling Data Cloud (optional smoke test)

■■ callout=true on @InvocableMethod

Both updated classes add callout=true to the @InvocableMethod annotation. This is required because ConnectApi.CdpQuery internally n

